Grounded Study Item Generation from Personal Notes

An Obsidian Plugin with a Multi-Agent Pipeline & Per-Concept Evaluation

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Problem

Personal note vaults (Obsidian, Notion, Logseq) accumulate **concept notes, PDFs, and figures**, but converting them into study items — problems, flashcards, code challenges — is manual, time-consuming, and error-prone.

LLMs can draft items cheaply but routinely **drift** from the exact content the learner stored, breaking instructional trust. Grading drift is worse: a single scalar score punishes *every concept* on a failed answer, including ones the student never even tried to use.

Goal

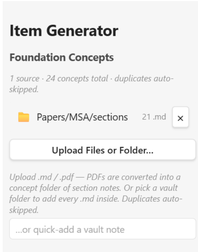
Generate study items grounded in the **learner's own notes**, grade them at the **concept level**, and never move raw notes outside the vault.

Approach

The vault is the source of truth. A TypeScript plugin sends curated JSON views of selected concepts to a FastAPI service that runs a multi-agent loop under tool-calling (Anthropic Claude Haiku 4.5). Outputs are written back to the same vault as markdown.

- **Sample Evaluator** pre-grades a learner-supplied reference — markdown *or* an image of an exam clip — and threads pedagogical notes into the Generator.
- **Core hardening loop** (Generator $\rightarrow$ Actor $\rightarrow$ Reflector) refines items until they survive simulated solving; a back-edge re-emits when difficulty $\geq$ hard.
- **Answer Evaluator** returns per-concept verdicts: `correct`, `alt_path`, `misunderstood`, `untested`.
- **Mastery + depth-aware scheduler**: per-concept Elo updates; mastered concepts graduate to integration items; `focus_depth` advances along a topological ordering produced by a link-architect over ingested PDF sections.

System Architecture



Vault concepts surfaced via folder upload — a single paper PDF expands into 21 section notes, all auto-deduplicated.

Two feedback cycles

1. Sample item evaluation (pre-round)

Sample (md / image) $\rightarrow$ Sample Evaluator $\dashrightarrow$ `analysis_notes` $\rightarrow$ Generator

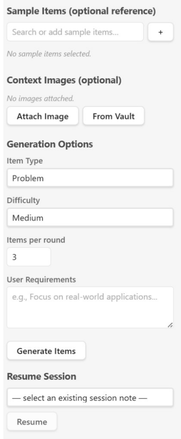
2. Core hardening loop (during round)

Generator $\rightarrow$ Actor$\times 3$ $\rightarrow$ Reflector $\dashrightarrow$ (if hard+) $\rightarrow$ Generator

Solid arrows are data flow, dashed arrows are feedback that conditions the *next* round. The Answer Evaluator (post-reply) closes a third loop on per-concept mastery and `focus_depth`.

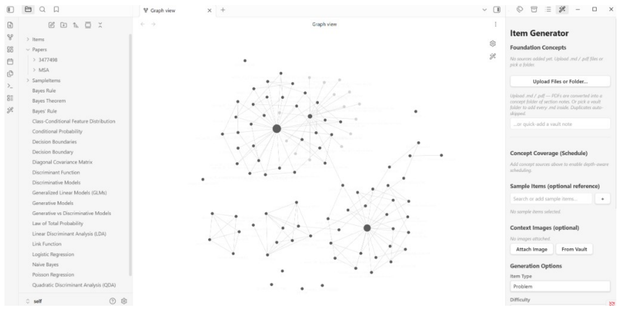
Why per-concept verdicts

Most LLM evaluators emit a scalar. We emit a **list of `ConceptVerdict` objects** so an alternate solution path still credits the concept it exercised, and a wrong answer never penalizes concepts the learner did not touch — directly feeding the Elo update.

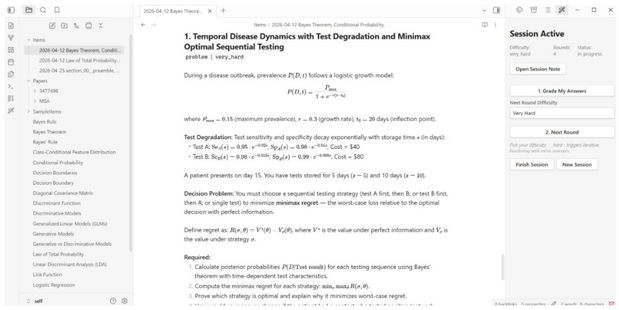


Generation options surfaced in the side panel: type, difficulty, items per round, free-text user requirements.

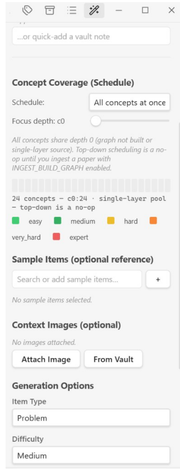
Plugin in Action



Obsidian global graph view — each concept becomes a node; the Item Generator panel reads selected sources directly from the vault.



A generated very_hard *problem combining Bayes' theorem with sequential testing under time-degrading sensitivity. LaTeX renders inline; the Session panel tracks rounds and exposes "Grade My Answers" / "Next Round" controls.*



Concept coverage row — each cell is one concept, colored by the highest difficulty at which it has been exercised; tick marks delineate dependency depths and the bolder tick marks `focus_depth`.

Pilot Evaluation (N=2)

Offline evaluation over the shipped `dataset.json` via the Braintrust harness:

- **91%** rubric satisfaction
- **69%** item clarity (LLM judge)
- **85%** structural validity
- **87.5%** latency band
- **100%** foundation coverage, prompt-shape, token-efficiency heuristics
- All cases completed without runtime errors

With $N{=}2$ these numbers **characterize the harness**, not a benchmark; no baselines reported. CIs are wide and unreported; means are descriptive only.

Post-pilot additions

Three features landed *after* the eval above was frozen — covered by unit tests but not yet in the quantitative table:

1. Per-concept **Answer Evaluator** + Elo mastery loop.
2. Multimodal **Sample Evaluator** (markdown or exam-clip image).
3. Depth-aware **scheduler** + concept coverage visualization.

Responsible AI

Raw notes stay in the vault. Cloud LLM and embedding calls are opt-in. The multimodal sample evaluator base64-encodes images in the browser before forwarding — a new exfiltration surface that is surfaced in-session only and deduplicated across evaluator and generator calls.

Limitations & Future Work

- In-memory sessions; not durable across restarts.
- Groq path is text-only; sample images stripped with a warning.
- MinerU PDF ingest brittle on multi-column layouts and low-quality scans.
- **Future:** stratified eval, human ratings, durable mastery storage, Groq multimodal, larger labeled trajectory set for the Answer Evaluator.

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